

Activité-type

Mail Server :

- Déploiement d'une infrastructure de messagerie sécurisée avec Mailcow, Postfix, Dovecot, SOGo et SendGrid sur VPS

1. Les tâches effectuées :

1. Objectif :

Mise en place d'un serveur de messagerie professionnel avec un domaine personnalisé afin d'établir une communication avec des services de messagerie externes ([Gmail](#), [Outlook](#), [Yahoo](#)) ...

2. Déploiement d'un serveur VPS, avec l'installation du système d'exploitation Ubuntu Server 22.04, afin d'héberger l'infrastructure de messagerie.

3. Configuration du nom de domaine personnel [zakirbabak.com](#) et du sous-domaine [mail.zakirbabak.com](#), gérés via [Cloudflare](#).

4. Déploiement de la solution de messagerie [Mailcow](#) à l'aide de Docker pour centraliser et automatiser les services.

4.1. Configuration de l'interface [webmail SOGo](#) pour l'accès utilisateur et la gestion des emails via navigateur.

5. Configuration des services de messagerie [Postfix](#) (SMTP) pour l'envoi des emails et [Dovecot](#) (IMAP) pour la réception.

6. Création et gestion des boîtes mail utilisateurs (ex : [info@zakirbabak.com](#) et [support@zakirbabak.com](#)) avec gestion des authentifications et des accès.

7. Intégration d'un relais SMTP externe via [SendGrid](#) afin d'améliorer la délivrabilité des emails vers les services externes ([Gmail](#), [Outlook](#), [Yahoo](#)).

8. Configuration des enregistrements [DNS](#) essentiels via [Cloudflare](#) pour sécuriser et authentifier les emails.

9. Mise en place du chiffrement SSL/TLS avec génération automatique de certificats Let's Encrypt via Mailcow pour sécuriser les communications.

10. Activation des connexions sécurisées SMTP (port 587 avec STARTTLS) et IMAP sécurisé (port 993).

11. Réalisation de tests d'envoi et de réception d'emails vers différents fournisseurs ([Gmail](#), [Outlook](#)) afin de valider le fonctionnement du système.

12. Analyse des logs des services Mailcow ([Postfix](#) et [Dovecot](#)) pour vérifier la bonne livraison et diagnostiquer les erreurs.

2. Environnement technique et moyens utilisés :

- VPS Hetzner : <https://www.hetzner.com/>
- Ubuntu Server 22.04
- Docker et Docker Compose
- Postfix, Dovecot, Mailcow et Sogo
- SendGrid (SMTP Relay)
- Cloudflare (gestion DNS et nom de domaine)
- Protocoles SMTP, IMAP et TLS/SSL
- Clients de messagerie : Gmail, Outlook et Webmail
- Documentation officielle Postfix et SendGrid, tutoriels YouTube et assistance IA (Claude)

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- https://www.postfix.org/documentation.html?utm_source=
- https://www.twilio.com/docs/sendgrid/for-developers/sending-email/getting-started-smtp?utm_source

3. Contexte

Projet réalisé de manière autonome dans le cadre de la formation professionnelle TSSR, en environnement cloud (VPS).

DNS management for zakirbabak.com

Review, add, and edit DNS records. Changes propagate globally within minutes.

DNS Setup: Full ⓘ [Import and Export](#) ▼

[Cloudflare Docs](#)

Search DNS Records

▼ Add filter

Q

Search

➕ Add record

	Domain	Status ⓘ
	zakirbabak.com	✓ Active

☐	Type ⓘ	▲ Name ⓘ	Content ⓘ	Proxy status ⓘ	TTL ⓘ	Actions
☐	CNAME	autodiscover	mail.zakirbabak.com	☁ DNS only	Auto	Edit ▶
☐	CNAME	autoconfig	mail.zakirbabak.com	☁ DNS only	Auto	Edit ▶
☐	TXT	_dmarc	"v=DMARC1; p=none"	DNS only	Auto	Edit ▶
☐	TXT	zakirbabak.com	"v=spf1 ip4: [REDACTED]"	DNS only	Auto	Edit ▼

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```
root@mail:~# cd /opt
root@mail:/opt# git clone https://github.com/mailcow/mailcow-dockerized
Cloning into 'mailcow-dockerized'...
remote: Enumerating objects: 59782, done.
remote: Counting objects: 100% (181/181), done.
remote: Compressing objects: 100% (85/85), done.
remote: Total 59782 (delta 135), reused 96 (delta 96), pack-reused 59601 (from 3)
Receiving objects: 100% (59782/59782), 48.99 MiB | 32.33 MiB/s, done.
Resolving deltas: 100% (39948/39948), done.
root@mail:/opt# cd mailcow-dockerized
root@mail:/opt/mailcow-dockerized# ./generate_config.sh
Detecting if your IP is listed on Spamhaus Bad ASN List...
Check completed! Your IP is clean
Press enter to confirm the detected value '[value]' where applicable or enter a custom value.
Mail server hostname (FQDN) - this is not your mail domain, but your mail server's hostname: mail.zakirbabak.com
Timezone [Etc/UTC]: Europe/Paris
Which branch of mailcow do you want to use?

Available Branches:
- master branch (stable updates) | default, recommended [1]
- nightly branch (unstable updates, testing) | not-production ready [2]
- legacy branch (supported until February 2026) | deprecated, security updates only [3]
Choose the Branch with it's number [1/2/3]
```

```
Copying snake-oil certificate...
root@mail:/opt/mailcow-dockerized# docker compose pull
[+] pull 143/189
* Image ghcr.io/mailcow/postfix-tlsp... [###] 22.1MB / 78.35MB Pulling 5.7s
* Image ghcr.io/mailcow/olefy:1.15 [###] 12.04MB / 48.7MB Pulling 5.7s
* Image ghcr.io/mailcow/phpfpm:8.2.29-2 [#####] 31.18MB / 100.4MB Pulling 5.7s
* Image mcuadros/ofelia:latest [####] 11.72MB / 12.32MB Pulling 5.7s
* Image ghcr.io/mailcow/rspamd:3.14.3-1 [###] 17.83MB / 79.33MB Pulling 5.7s
* Image ghcr.io/mailcow/unbound:1.25 [#####] 13.3MB / 36.36MB Pulling 5.7s
* Image mariadb:10.11 [###] 14.5MB / 111.3MB Pulling 5.7s
* Image redis:7.4.6-alpine [#####] 8.045MB / 16.26MB Pulling 5.7s
* Image ghcr.io/mailcow/postfix:3.7.11-2 [###] 17.84MB / 110MB Pulling 5.7s
* Image ghcr.io/mailcow/clamd:1.71 [#####] 18.88MB / 36.62MB Pulling 5.7s
* Image ghcr.io/mailcow/dockerapi:2.12 [###] 12.26MB / 50.58MB Pulling 5.7s
* Image ghcr.io/mailcow/nginx:1.06 [#####] 20.41MB / 49.03MB Pulling 5.7s
* Image ghcr.io/mailcow/watchdog:2.11 [#####] 13.31MB / 39.51MB Pulling 5.7s
* Image ghcr.io/mailcow/dovecot:2.3.... [#####] 10.1MB / 110.2MB Pulling 5.7s
* Image memcached:alpine [#####] 5.98MB / 5.98MB Pulling 5.7s
* Image ghcr.io/mailcow/acme:1.97 [###] 13.32MB / 42.72MB Pulling 5.7s
* Image ghcr.io/mailcow/netfilter:1.64 [#####] 13.31MB / 44.08MB Pulling 5.7s
* Image ghcr.io/mailcow/sogo:5.12.5-3 [#####] 18.89MB / 172.6MB Pulling 5.7s
```

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```
root@mail:/opt/mailcow-dockerized# docker compose exec mysql-mailcow "USE mailcow; SELECT domain, active FROM domain;"
+-----+-----+
| domain      | active |
+-----+-----+
| zakirbabak.com |      1 |
+-----+-----+
```



Postfix Configuration

The 'mail name' is the domain name used to 'qualify' _ALL_ mail addresses without a domain name. This includes mail to and from <root>: please do not make your machine send out mail from root@example.org unless root@example.org has told you to.

This name will also be used by other programs. It should be the single, fully qualified domain name (FQDN).

Thus, if a mail address on the local host is foo@example.org, the correct value for this option would be example.org.

System mail name:

zakirbabak.com

<Ok> <Cancel>

```
root@mail:/opt/mailcow-dockerized# systemctl status postfix
● postfix.service - Postfix Mail Transport Agent
   Loaded: loaded (/lib/systemd/system/postfix.service; enabled; vendor preset: enabled)
   Active: active (exited) since Wed 2026-04-22 13:05:43 UTC; 5min ago
     Docs: man:postfix(1)
   Main PID: 8820 (code=exited, status=0/SUCCESS)
    CPU: 861us
```

```
root@mail:/opt/mailcow-dockerized# docker compose restart sogo-mailcow
[+] restart 0/1
  ✨ Container mailcowdockerized-sogo-mailcow-1 Restarting
root@mail:/opt/mailcow-dockerized# docker compose restart acme-mailcow
[+] restart 0/1
  ✨ Container mailcowdockerized-acme-mailcow-1 Restarting
root@mail:/opt/mailcow-dockerized#
```

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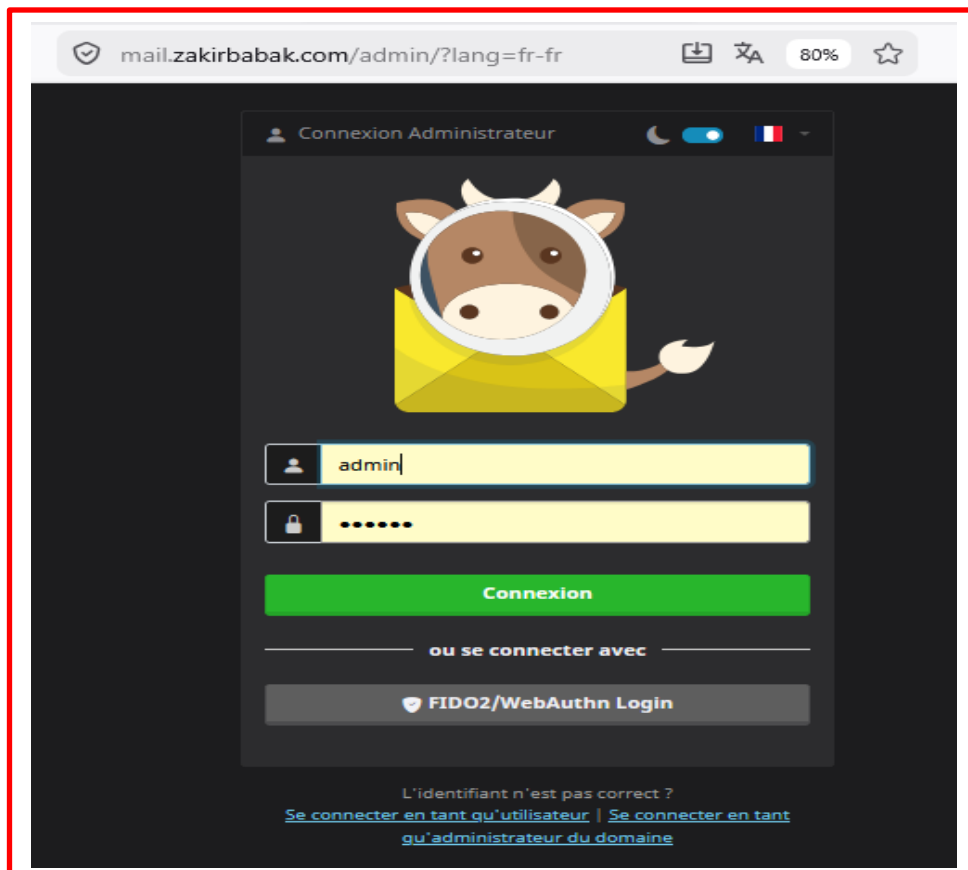
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```
root@mail:~# sudo hostnamectl set-hostname mail.zakirbabak.com
root@mail:~# sudo nano /etc/hosts
root@mail:~# nslookup mail.zakirbabak.com
Server:         127.0.0.53
Address:        127.0.0.53#53

Name:   mail.zakirbabak.com
Address: 127.0.1.1

root@mail:~# hostname
mail.zakirbabak.com
root@mail:~# hostnamectl
  Static hostname: mail.zakirbabak.com
            Icon name: computer-vm
            Chassis: vm
            Machine ID: [REDACTED]
            Boot ID: [REDACTED]
  Virtualization: kvm
  Operating System: Ubuntu 22.04.5 LTS
            Kernel: Linux 5.15.0-176-generic
  Architecture: x86-64
  Hardware Vendor: Hetzner
  Hardware Model: vServer
```



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Add mailbox

Username (left part of an email address)

info

Full name

Info Mailbox

Domain

zakirbabak.com

Template

Default

Password (generate)

.....

Check against haveibeenpwned.com

- Minimum password length is 6

Confirmation password (repeat)

.....

Tags

Quota (MiB)

1024

max. 10240 MiB

0 = ∞

Set handling for tagged mail

In subfolder

In subject

Do nothing

In subfolder: a new subfolder named after the tag will be created below INBOX ("INBOX/Facebook").

In subject: the tags name will be prepended to the mails subject, example: "[Facebook] My News".

Example for a tagged email address: me+Facebook@example.org

Quarantine notifications

Never

Hourly

Daily

Weekly

Once a notification has been sent, items will be marked as "notified" and no further notifications will be sent for this particular item.

Quarantine notification category

Rejected

Junk folder

All categories

The notification category "Rejected" includes mail that was rejected, while "Junk folder" will notify a user about mails that were put into the junk

Encryption policy

Enforce TLS incoming

Enforce TLS outgoing

Allowed protocols for direct user access (does not affect app password protocols)

IMAP, POP3, SMTP, Sieve

ACL

Temporary aliases, TLS policy, Spam score, De

Rate limit

Disabled

msgs / second

This rate limit is applied on the SASL login name, it matches any "from" address used by the logged-in user. A mailbox rate limit is a domain-wide rate limit.

Active

☐ Force password update at next login

This user will only be able to login to mail.zakirbabak.com - mail UI. App passwords remain useable.

☐ Force 2FA enrollment at login

The user will be required to set up two-factor authentication before accessing the panel.

☒ Direct forwarding to SOGo

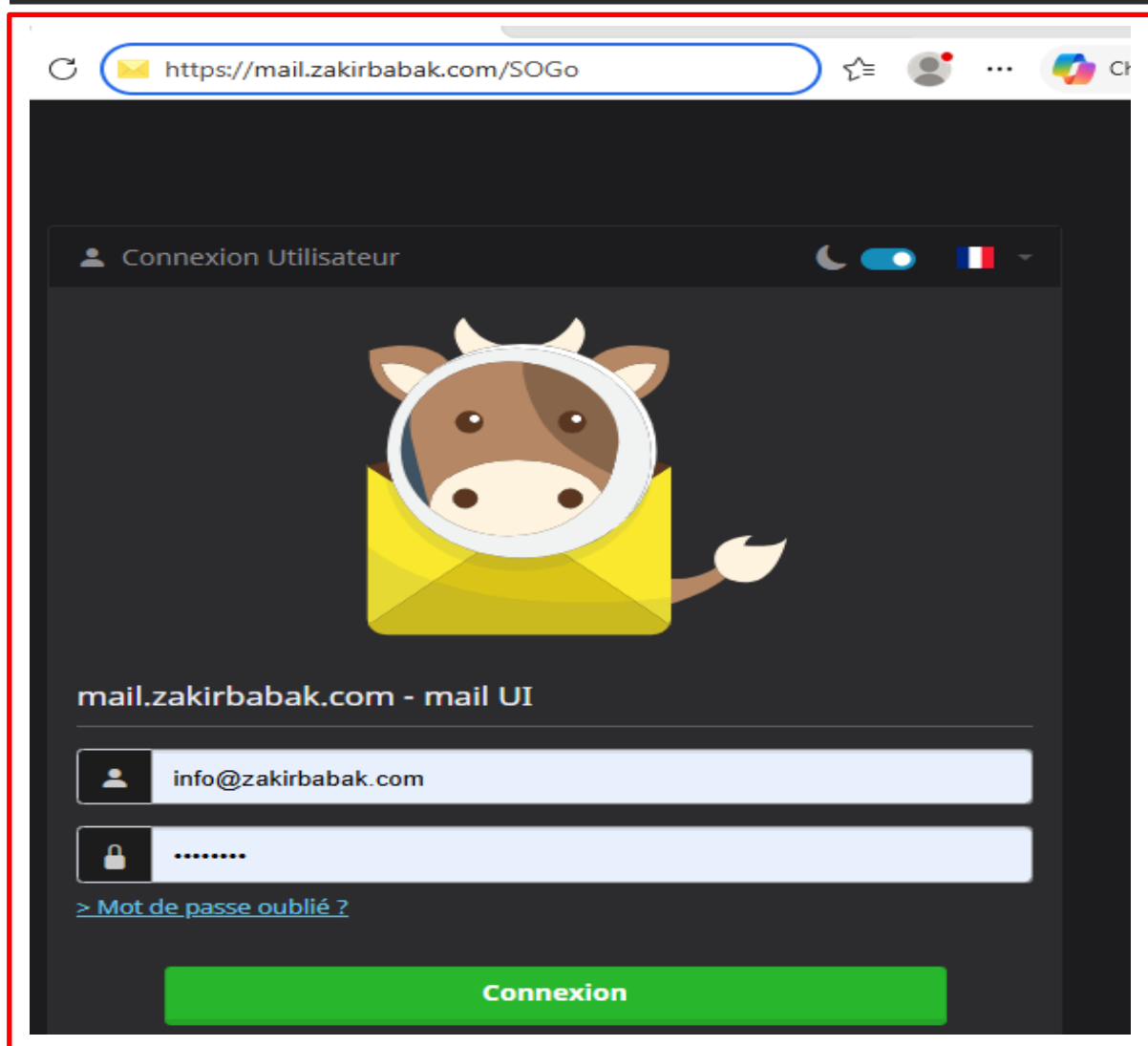
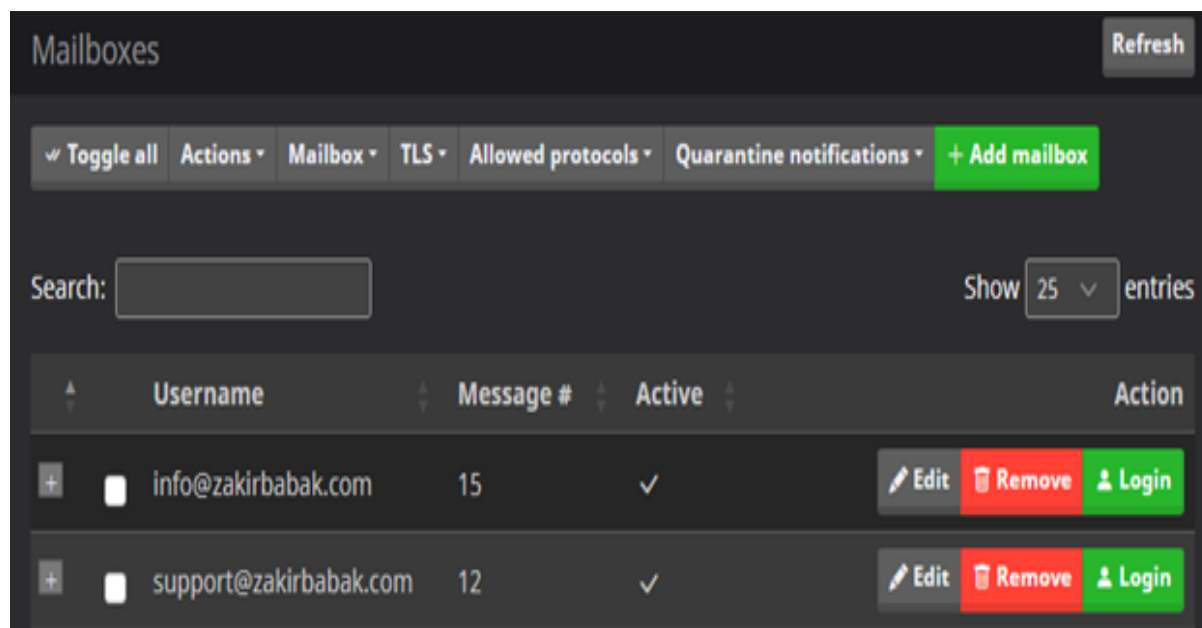
After logging in, the user is automatically redirected to SOGo.

Add

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Integrate using our Web API or SMTP Relay



✓ Overview

2 Integrate

1

Create an API key

This allows your application to authenticate to our API and send mail. You can enable or disable additional permissions on the [API keys page](#).



✓

"mailcow-sendgrid-relay" was successfully created and added to the next step.

Bonjour Outlook, Gmail et Postfix Inbox

 **info@zakirbabak....** 01:49 
to zakirbabk, me, support ^

From Info Mailbox info@zakirbabak.com
To @outlook.com
@gmail.com
support@zakirbabak.com

Date May 5, 2026 at 01:49

 Standard encryption (TLS)
[Learn more](#)

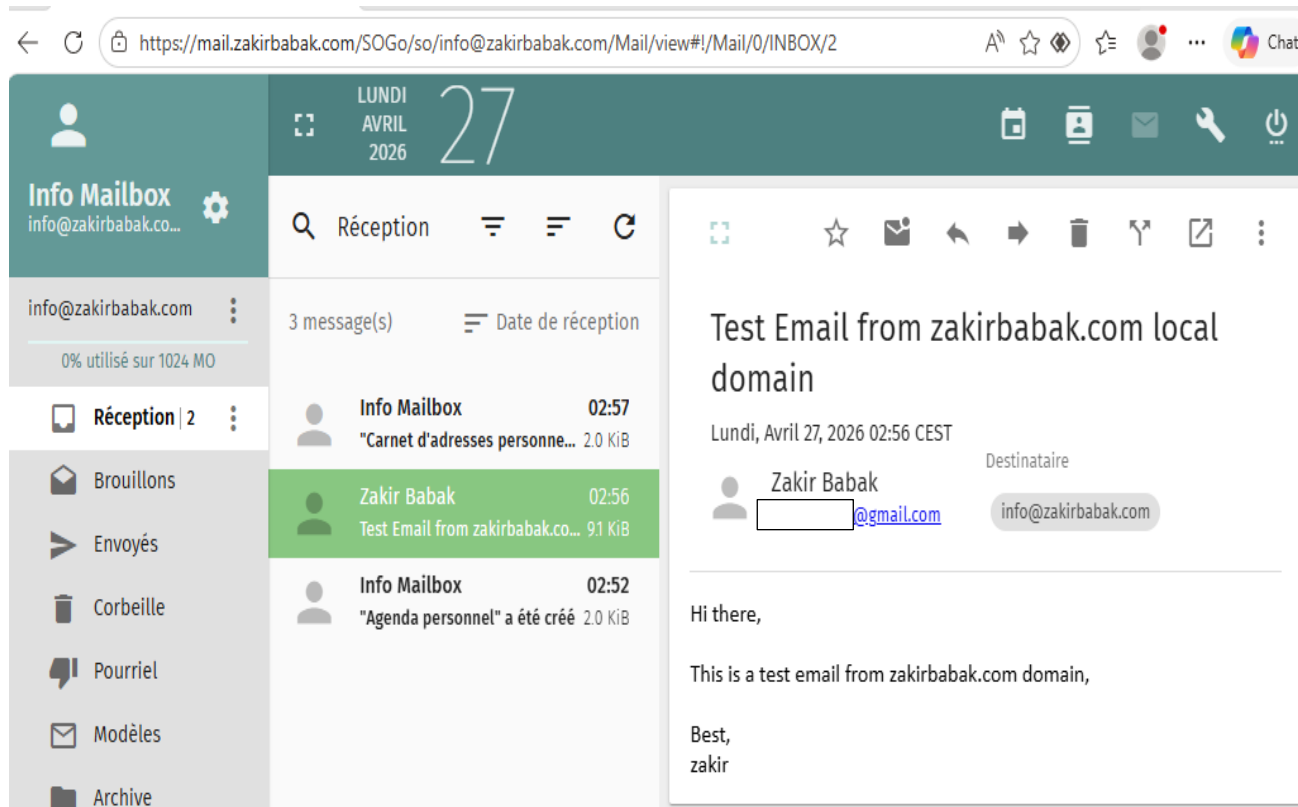
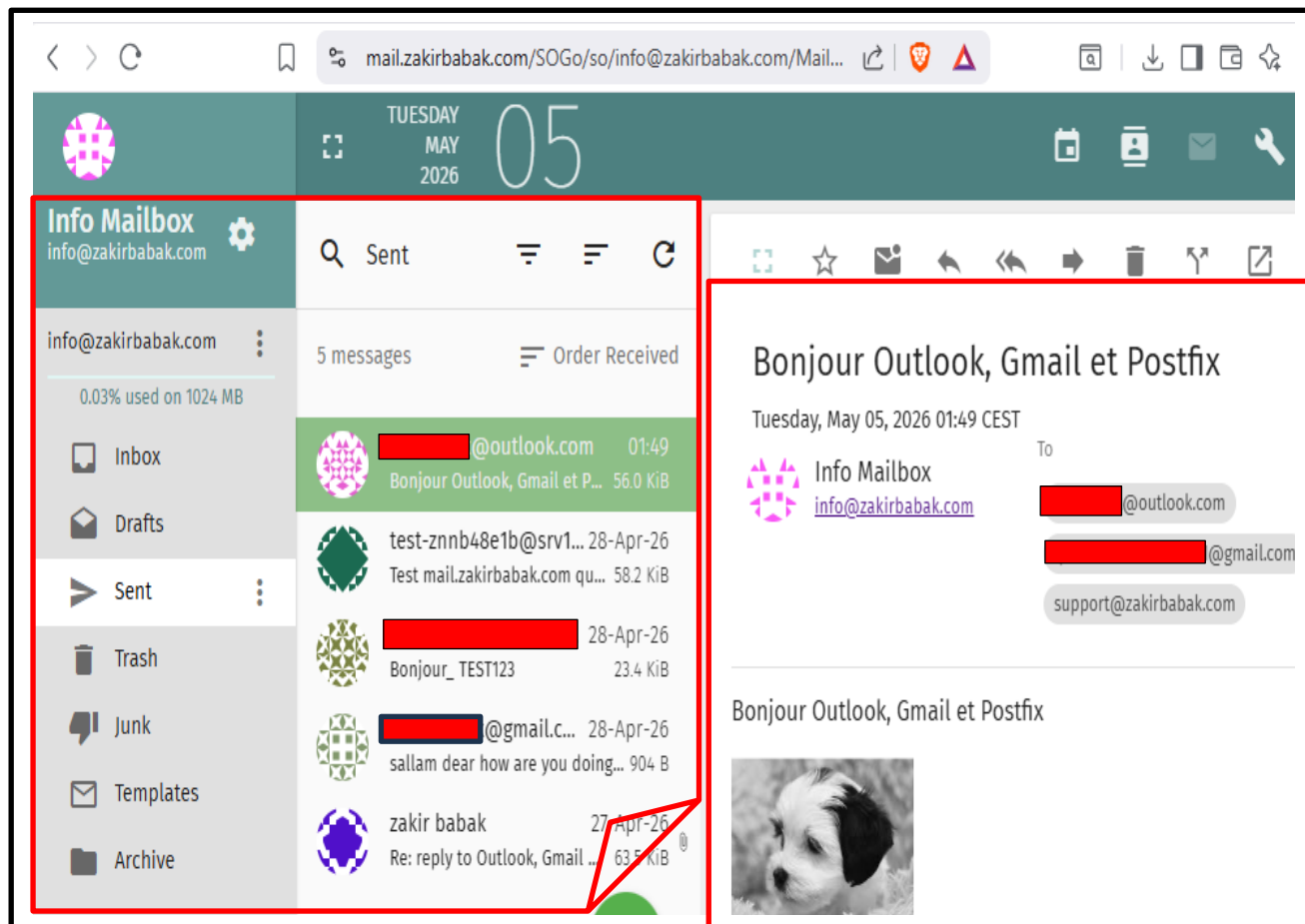
Bonjour Outlook, Gmail et Postfix



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https://mail.zakirbabak.com/SOGo/so/info@zakirbabak.com/Mail/view#!Mail/0/INBOX/6

Inbox

6 messages

Order Received

support mailbox	03:52
Re: reply to Outlook, Gmail & MailCo...	61.8 KiB
zakir babak	03:30
This email from outlook	13.6 KiB
zakir babak	03:11
Response to test email	8.2 KiB
Info Mailbox	02:57
"Carnet d'adresses personnel" a été c...	2.0 KiB
Zakir Babak	02:56
Test Email from zakirbabak.com local ...	9.1 KiB
Info Mailbox	02:52
"Agenda personnel" a été créé	2.0 KiB

Re: reply to Outlook, Gmail & MailCow domains

Monday, April 27, 2026 03:52 CEST

To: support mailbox
support@zakirbabak.com

Cc: Info Mailbox

zakir babak [redacted]@gmail.com

What a lovely puppy here!

On Monday, April 27, 2026 03:46 CEST, "Info Mailbox" <info@zakirbabak.com> wrote.

Hey Outlook and Gmail, your messages have been received well; this is a confirmation.



On Monday, April 27, 2026 03:30 CEST, zakir babak [redacted]@outlook.com> wrote:

This is a test email from Outlook, make sure you get it.

Send from iPhone

Sent from [Outlook for iOS](#)

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https://mail.zakirbabak.com/SOGo/so/info@zakirbabak.com/Mail/view#/Mail/0/INBOX/6

MONDAY APRIL 27 2026

Info Mailbox info@zakirbabak.co...

info@zakirbabak.com 0.16% used on 1024 MB

Inbox | 3

Drafts

Sent

Trash

Junk

Templates

Archive

9 messages Order Received

support mailbox	04:07	820 KiB
Re: image send test		
zakir babak	04:03	760 KiB
Re: From Gmail on iPhone		
zakir babak	04:03	7.8 KiB
From Gmail on iPhone		
support mailbox	03:52	61.8 KiB
Re: reply to Outlook, Gmail ...		
zakir babak	03:30	13.6 KiB
This email from outlook		
zakir babak	03:11	8.2 KiB
Response to test email		
Info Mailbox	02:57	2.0 KiB
"Carnet d'adresses personnel...		
Zakir Babak	02:56	9.1 KiB
Test Email from zakirbabak.co...		
Info Mailbox	02:52	2.0 KiB
"Agenda personnel" a été créé		

Re: reply to Outlook, Gmail & MailCow domains

Monday, April 27, 2026 03:52 CEST

To support mailbox support@zakirbabak.com

Cc zakir babak [REDACTED]@gmail.com

What a lovely puppy here!

On Monday, April 27, 2026 03:46 CEST, "Info Mailbox" <info@zakirbabak.com> wrote.

Hey Outlook and Gmail, your messages have been received well; this is a confirmation.



Re: image send test

Monday, April 27, 2026 04:07 CEST

To support mailbox support@zakirbabak.com

Cc zakir babak info@zakirbabak.com



On Monday, April 27, 2026 04:03 CEST, zakir babak [REDACTED]@gmail.com> wrote:

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app.sendgrid.com/email_activity?filters=""&isAndOperator=true&page=1

twilio SendGrid My SendGrid Account Trial: [API & MC Upgrade](#)

Activity Feed

To email address 2026/04/25 - 2026/04/28 Clear

STATUS	MESSAGE	LAST EVENT RECEIVED	OPENS
Delivered	To: test-znnb48e1b@srv1.mail-tester.com Test mail.zakirbabak.com quality of service	2026/04/28 12:04am UTC+00:00	1
Delivered	To: [REDACTED]@gmail.com Re: Test Email from zakirbabak.com local do...	2026/04/27 11:48pm UTC+00:00	5
Delivered	To: [REDACTED]@gmail.com Re: reply to Outlook, Gmail & MailCow domains	2026/04/27 11:48pm UTC+00:00	4
Delivered	To: [REDACTED]@gmail.com Test Direct_CCC_BCC_Message	2026/04/27 11:48pm UTC+00:00	2
Delivered	To: [REDACTED]@gmail.com Test Direct_CCC_BCC_Message	2026/04/27 11:48pm UTC+00:00	3

Reputation 93%
VIEW ACCOUNT USAGE

6 messages 191.73

0%

Protocols: ? IMAP ✓ SMTP ✓ Sieve ✓ POP3 ✓ ActiveSync ✗ CalDAV/CardDAV ✗

Apple connection profile: ? Email [IMAP, SMTP]
? Email, calendars and contacts [IMAP, SMTP, Cal/CardDAV]

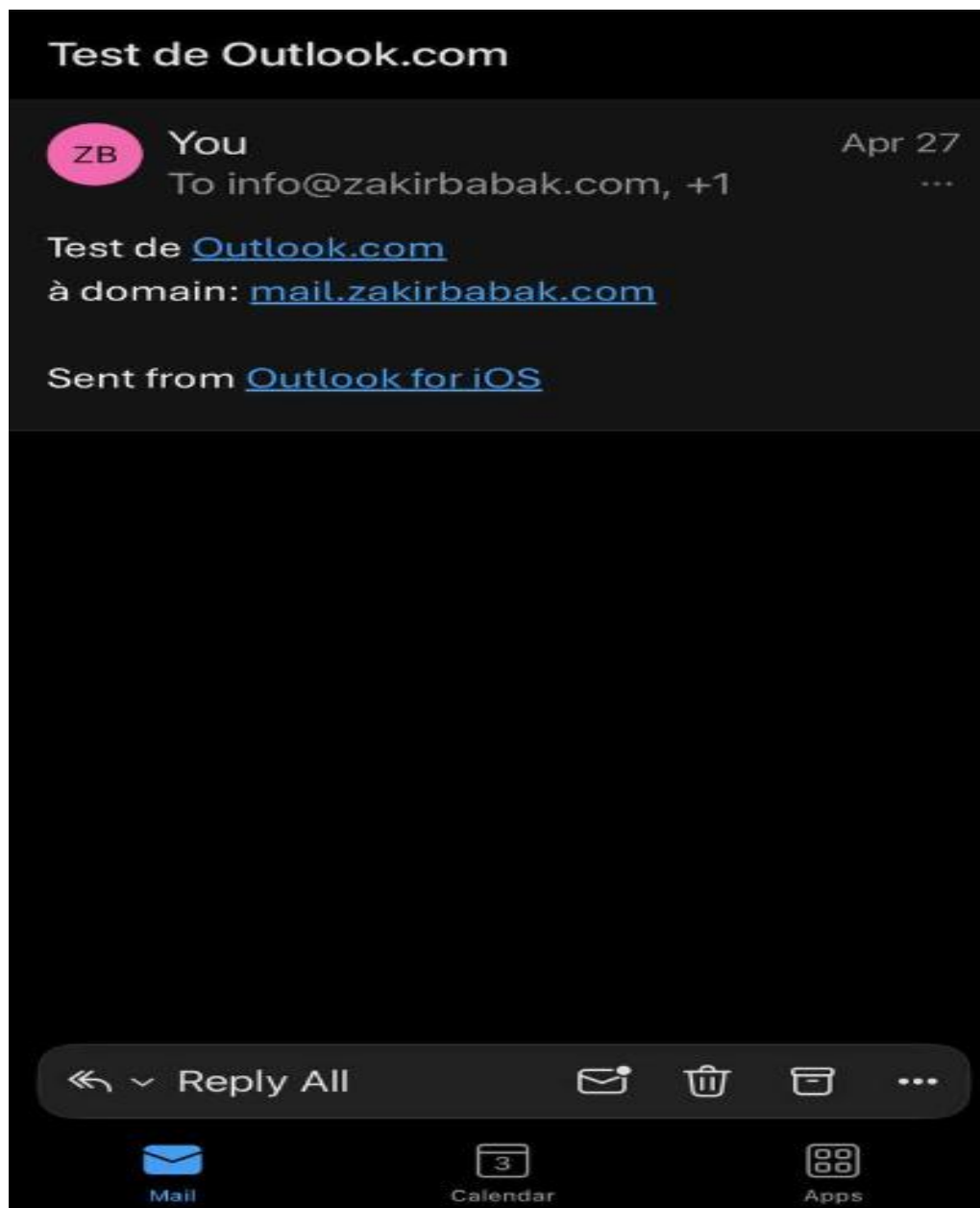
Apple connection profile with app password: ? Email [IMAP, SMTP]
? Email, calendars and contacts [IMAP, SMTP, Cal/CardDAV]

[Show configuration guides for email clients and smartphones](#)
[Show active user sieve filter](#)

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DNS Records			
Please note that changes made to DNS may take up to 24 hours to correctly have their current state reflected on this page. It is intended as a way for you to easily see how to configure your DNS records and to check whether all your records are correctly stored in DNS.			
Name	Type	Correct Data	Current State
mail.zakirbabak.com	A		✓
	PTR	mail.zakirbabak.com	static s.your-server.de
mail.zakirbabak.com	AAAA	2a01:04f8:1c19:333f:0000:0000:0000:0001	2
	PTR	mail.zakirbabak.com	2
_25._tcp.mail.zakirbabak.com	TLSA		2
zakirbabak.com	MX	mail.zakirbabak.com	✓
autodiscover.zakirbabak.com	CNAME	mail.zakirbabak.com	✓
_autodiscover._tcp.zakirbabak.com	SRV	mail.zakirbabak.com 443	2
autoconfig.zakirbabak.com	CNAME	mail.zakirbabak.com	✓
zakirbabak.com	TXT	SPF Record Syntax	? v=spf1 ip4: all ^
_dmarc.zakirbabak.com	TXT	DMARC Assistant	v=DMARC1;p=none ^

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```
root@mail:~# ufw status numbered
Status: active

      To Action From
      --
[ 1] 80/tcp ALLOW IN Anywhere
[ 2] 443/tcp ALLOW IN Anywhere
[ 3] 25/tcp ALLOW IN Anywhere
[ 4] [REDACTED] ALLOW IN Anywhere
[ 5] 587/tcp ALLOW IN Anywhere
[ 6] 993/tcp ALLOW IN Anywhere
[ 7] [REDACTED] ALLOW IN Anywhere
[ 8] 143/tcp ALLOW IN Anywhere
[ 9] 465/tcp ALLOW IN Anywhere
[10] 80/tcp (v6) ALLOW IN Anywhere (v6)
[11] 443/tcp (v6) ALLOW IN Anywhere (v6)
[12] 25/tcp (v6) ALLOW IN Anywhere (v6)
[13] [REDACTED] (v6) ALLOW IN Anywhere (v6)
[14] 587/tcp (v6) ALLOW IN Anywhere (v6)
[15] 993/tcp (v6) ALLOW IN Anywhere (v6)
[16] [REDACTED] (v6) ALLOW IN Anywhere (v6)
[17] 143/tcp (v6) ALLOW IN Anywhere (v6)
[18] 465/tcp (v6) ALLOW IN Anywhere (v6)
```